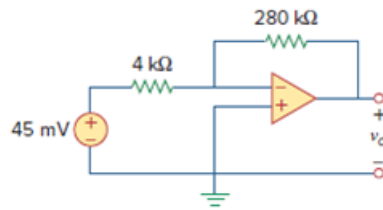


Problem

Find the output of the op amp circuit shown in Fig. 5.13. Calculate the current through the feedback resistor.

Figure. 5.13:



Step-by-step solution

Step 1 of 4

Draw the labeled circuit diagram.

Step 2 of 4

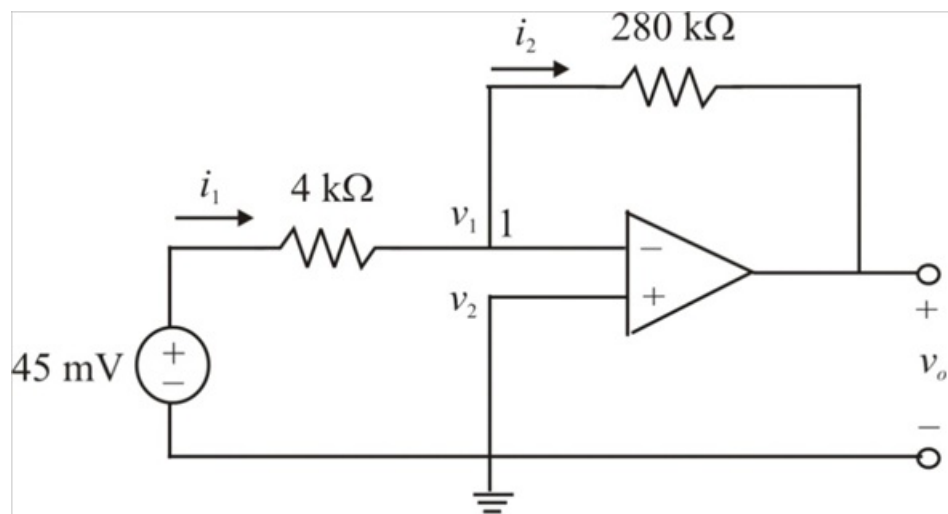


Figure 1

Step 3 of 4

Applying Kirchhoff's current law at node 1.

For an ideal opamp, the input current inside the opamp will be zero. So,

$$i_1 = i_2$$
$$\frac{(45 \times 10^{-3}) - v_1}{4 \text{ k}\Omega} = \frac{v_1 - v_o}{280 \text{ k}\Omega}$$

But $v_1 = v_2 = 0$ for an ideal op amp, since non-inverting terminal is grounded. Hence,

$$\frac{(45 \times 10^{-3}) - 0}{4 \text{ k}\Omega} = \frac{0 - v_o}{280 \text{ k}\Omega}$$

$$\frac{(45 \times 10^{-3})}{4} = \frac{-v_o}{280}$$

$$v_o = \frac{-(45 \times 10^{-3})(280)}{4}$$

$$v_o = -3.15 \text{ V}$$

Therefore, the output voltage v_o is $\boxed{-3.15 \text{ V}}$.

Step 4 of 4

Calculate the current through the feedback resistor.

$$i_2 = \frac{v_1 - v_o}{R_f}$$
$$= \frac{0 - (-3.15)}{280 \times 10^3}$$
$$= 11.25 \mu\text{A}$$

Therefore, the current through the feedback resistor is $\boxed{11.25 \mu\text{A}}$.